

GIUSEPPE FESTA

Control and AI Engineer for Humanoid Robots and Autonomous Systems

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Control and embedded systems engineer specializing in real-time software for humanoid robots and autonomous platforms. Experience spans the full robotics stack from PCB and firmware to ROS2 control and perception across Neura Robotics, Michelin, and Terna, plus two co-founded startups.

Currently leading embedded control development for humanoid robots in Zürich.

Always seeking an opportunity to change the world.

PROFESSIONAL EXPERIENCE

- Neura Robotics | Zurich, Switzerland**
Embedded Control Engineer Team Lead for Humanoid Robots 2025-Present
- Lead a team of 2 engineers developing real-time embedded systems for humanoid locomotion, owning the stack from firmware up to the DDS communication layer. Custom application development for robotic control systems and state estimation.
 - Designed and deployed RT control software on embedded platforms, meeting sub-millisecond latency requirements for joint-level control loops.
- Control Engineer for Humanoid Robots Metzingen, Germany 2025**
- Designed a low level control system and interface, paired with a RT communication stack for robotic actuation.
 - Developed and maintained the low latency DDS infrastructure to support humanoid robot peripherals and sensors
- Thinking! | Stage Robotics Milan, Italy - Lille, France**
Robotics Engineer 2022-2025
- Design of a custom robot, able to perform unique movements on the stage. The realisation includes 3D modelling and printing, Python and ROS2 software development, wiring, and RF coupling via Altium and Espressif IDE.
 - Design of Synchronised Systems and Firmware interfaces with DMX512 to control distributed systems during shows and live exhibitions, and production of documentation to support safety standards with Italian V.V.F.F. and French DCCRS.
- Michelin | Cuneo - Fossano Plant Cuneo, Italy**
Perception Engineer 2022-2024
- Successfully developing a fast and reliable fault detection system to improve online quality controls on the Michelin car tire production facility. The job required extensive Machine Learning skills, deep knowledge of OpenCV and TensorFlow libraries for C++ and Python, and optimization done through Vulcan for workload parallelisation.
- FreeRoom (freeroom.digital) Milan, Italy - Valencia, Spain**
Co-Founder and CTO 2022-2024
- Founded a startup and implemented the perception and secure networking for meshed control cameras. The enterprise has since grown by acquiring partners interested in utilising advanced machine vision for security purposes.
- Dynamis PCR | PoliMi Reparto Corse Milan, Italy**
Inverter Designer 2020-2023
- In a team of three people, I successfully developed a three-phase 20 kW modular inverter to drive custom motors. Designed and developed PCB and custom FOC software using MATLAB, OrCAD, CubeMX and STM32 Keil.

EDUCATION

- Politecnico di Milano Milan, Italy**
Msc. Automation and Control Engineering 2022-2024
- GPA:** 4.00 - 110 Cum Laude
 - Thesis:** Terna S.p.A.: *Design, Implementation and Experimental Tests of a Prototype of Power Converters for HVDC Applications.*
- Politecnico di Milano Milan, Italy**
Bsc. Automation and Control Engineering 2019-2022
- GPA:** 4.00 - 110
 - Relevant Activities:** Proven track record in Electronics, Machine Learning, Computer Vision, and Control Applications.
- Alta Scuola Politecnica Milan - Turin, Italy**
- GPA:** 4.00
 - Completed the prestigious ASP honour program (PoliMi & PoliTo) focused on excellence in science and engineering.
- RWTH Aachen Aachen, Germany**
Erasmus+ Student 2023-2024
- FAU Erlangen - Siemens Additive Manufacturing Nuremberg, Germany**
Additive Manufacturing and Powders Processing Techniques Trainee. 2023-2024
- Università Parthenope di Napoli Naples, Italy**
CyberChallenge Cyber Security Training Program. Head of the Regional Team for the national phase. 2018 – 2020

ACTIVITIES & LEADERSHIP

- Athens Program Ambassador - Politechnika Warszawska Warsaw, Poland**
- Head of the team at the Warsaw University of Technology for the training program “Advanced Acoustic Measurements”.

SKILLS & INTERESTS

Interests: Boxing, MMA, Guitar, Rugby, Climbing, 3D Modeling, Robotics, ML
Languages: Italian (native), English (C2), Spanish (B2), German (A2), Turkish (A2)